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Transparent information fusion network: An explainable network for multi-source bearing fault diagnosis via self-organized neural-symbolic nodes

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ABSTRACT

In recent years, the integration of Artificial Intelligence (AI) into Intelligent Fault Diagnosis (IFD) through multi-source signal fusion has advanced significantly. However, the inherent opacity of AI-driven IFD models often hampers explainability, a crucial factor for fostering deeper collaboration between humans and intelligent systems to enhance the safety and reliability of industrial assets. This study introduces a novel and explainable methodology termed the Transparent Information Fusion Network (TIFN). TIFN incorporates multiple self-organized Neural-Symbolic Nodes (NSNs) equipped with signal processing and statistical operators, as opposed to conventional black-box neural networks or manually crafted expert systems. NSNs are interconnected through learnable signal-wise gates to construct the Signal Operator Layer (SOL) and Feature Operator Layer (FOL). The entire network is trainable using gradient-based learning methods in a self-organized manner. This enables accurate representation of fault signals and establishes a fully comprehensible semantic framework with an explainable decision-making process. The transparency, generalization, and capability to learn from limited samples of TIFN are demonstrated through two case studies on rotating machinery equipped with different sensors. Case 1 fuses vibration and piezoelectric signals, while Case 2 integrates piezoelectric and triboelectric signals to achieve comprehensive information fusion at both signal and feature levels. By incorporating learnable NSNs, signal-wise gates, TIFN achieves superior diagnostic performance with fewer parameters compared to traditional expert-organized models. This research underscores the potential of TIFN as a fully explainable tool for industrial diagnostics with multi-source signals, paving the way for enhanced human-machine collaboration in Industry 5.0, with a focus on trustworthiness, transparency, and accountability.

1. Introduction

AI-based methodologies assist manufacturers in Industry 4.0 in predicting maintenance needs, ushering in a new era where digital technologies powered intelligent and automated systems [1]. Central to these advances is the development of Intelligent Fault Diagnosis (IFD), which is crucial for maintaining the efficiency, reliability, and safety of industrial operations [2]. IFD systems harness the power of advanced sensors, data analytics, and machine learning to predict and diagnose equipment malfunctions before they lead to catastrophic failures [3]. Therefore, IFD has gained significant importance in minimizing operational downtime in industries [4,5] and serves as pivotal components in the predictive maintenance strategies of modern industries [6].

The rapid advancement of modern AI has led to the emergence of numerous sophisticated intelligent information fusion techniques [7]. These cutting-edge methods have significantly inspired the IFD community, which extensively leverage information fusion to integrate data from multiple sources and enhance the accuracy and reliability of fault diagnosis [8]. This integration is crucial due to the complex working environments of mechanical systems [9]. On the one hand, it is necessary to conduct information fusion to synthesize information across multiple sensor inputs and analytical levels, providing a comprehensive understanding of machinery health and operational anomalies [10]. On the other hand, this multi-sensor approach is instrumental in detecting subtle anomalies that might be missed when monitoring data from a single source [11].

In the field of IFD, there are three levels of information fusion [12]: data, feature, and decision levels. Fusion at the data level involves

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merging raw data from various sensors before additional processing or analysis, maintaining sensor data integrity and resolution to establish a solid foundation for subsequent diagnostic algorithms. For example, Sun et al. [13] proposed a non-contact sensing technology to explore the fusion mechanism of multi-sensor heterogeneous measurements, including infrared thermal and acoustic data. Similarly, Li et al. [14] presented a composite neuro-fuzzy system-guided cross-modal zero-sample diagnostic framework by using infrared thermography and acoustic data to monitor gearbox conditions. Huang et al. [15] proposed empirical mode decomposition-based multi-wavelet packet energy entropy using multi-source three-dimensional blade tip clearance signals for turbine blades health maintaining. Zhao et al. [16] utilized various modal data such as image, sound, and vibration signals of the simulated wind turbine to introduce a multimodal double-layer detection system. Jiang et al. [17] introduced a semi-supervised optimal temperature-guided prototypical contrastive learning framework enabled by multi-sensor data fusion for machinery fault diagnosis in diverse working conditions with minimal annotated data. Li et al. [18] proposed a universal method compatible with different devices and multimodal sensing for industrial robots [19]. This method considers its suitability under various working conditions and employs a joint distribution adaptation technique based on lightweight networks. Feature level fusion occurs after the initial data processing stage, amalgamating pertinent features from various data origins to create a holistic feature set, thus decreasing data dimension while retaining vital fault diagnosis information. For instance, Liu et al. [20] proposed a multi-scale kernel algorithm with residual learning-enabled convolution neural network (CNN) architecture. Wan et al. [21] proposed a multi-sensor information coupling network that utilizes the deeper features from multi-sensors independently and simultaneously. Gao et al. [22] proposed a multi-source domain information fusion network aimed at obtaining comprehensive knowledge by abundant feature information. Chen et al. [23] proposed a novel continuous feature fusion technique called Lifelong Learning Method for Fault Diagnosis (LLMFD). The core of LLMFD is the Dual-Branch Aggregation Networks (DBANets) framework, which integrates reserved exemplars to effectively learn new fault types. Decision-level fusion combines the results from various sensors or models to make a final decision, enhancing the reliability and accuracy of fault diagnosis. Shao et al. [12] proposed a novel approach to multi-sensor fusion to diagnose faults in a collaborative manner with a stacked wavelet auto-encoder and a flexible weighted assignment of fusion strategies. Therefore,each level of information fusion contributes uniquely to the robustness and accuracy of IFD systems, supporting the complex requirements of modern industrial operations.

Unlike the IFD in Industry 4.0, which primarily focuses on reducing human labor through automation, Industry 5.0 emphasizes human-centric, sustainable, and resilient approaches [24,25]. While Industry 4.0 prioritizes the feasibility and implementation of technology, it often overlooks human needs. In contrast, the vision of Industry 5.0 places workers at the center of the process, positioning them as the actual decision-makers. This shift fosters deeper collaboration between humans and intelligent systems, aiming to optimize operational efficiency and minimize downtime [26]. Furthermore, technicians and managers require insights that elucidate the actions and rationale of intelligent systems, ensuring that fault predictions are contextualized and easily comprehensible to humans [27].

However, many AI-enabled IFD systems still rely on deep learning models that function as "black boxes". While these models are powerful, they inherently lack transparency and trustworthiness, making it difficult to understand how conclusions are drawn [28,29]. Consequently, Explainable Predictive Maintenance (XPM), defined as data-driven predictive maintenance using Explainable Artificial Intelligence (XAI), has gained significant momentum [30,31]. XPM is essential not only for identifying failures but also for elucidating the rationale behind these predictions [30,32,33]. Such explainability is crucial for planning maintenance activities and optimizing operations [34].

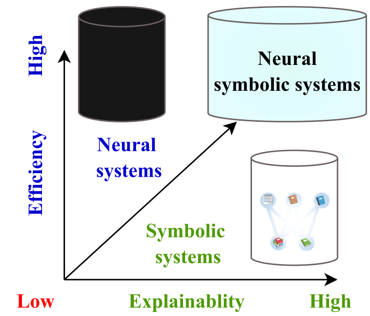

Fig. 1. Neural-symbolic learning systems.

To address the explainability challenge in information fusion for IFD, we introduce the concept of Transparent Fault Diagnosis (TFD) to pursue a fully understandable learning and diagnosis procedure. TFD is inspired by Neural-Symbolic AI (NSAI) [35], which effectively combines the powerful representation capabilities of neural systems with the explainability of symbolic systems, as illustrated in Fig. 1 [36,37]. NSAI leverages neural systems for machine learning tasks while incorporating symbolic knowledge into the training process to enhance performance and explainability [38]. For instance, Hogea et al. proposed LogicLSTM, which combines the strengths of LSTM deep recurrent networks for temporal data processing with logical reasoning capabilities, improving prediction accuracy and interpretability [39]. Li et al. [40] introduced the Deep Expert Network (DEN), which integrates expert knowledge via neuro-symbolic AI to achieve transparent decision-making in fault diagnosis. However, the development of NSAI in TFD is still in its infancy.

Despite the growing interest in TFD, several research gaps remain:

- **Opacity of deep learning models:** The inherent "black-box" nature of deep learning algorithms limits their explainability, making it difficult to understand how these models reach their conclusions.
- **Explainable expert knowledge fusing:** Achieving deeper collaboration between humans and intelligent systems remains challenging. Traditional black-box neural networks or manually crafted expert systems struggle to organize and utilize expert knowledge in a flexible and transparent manner, complicating the combination of AI insights with human expertise in decision-making processes.

To bridge the gap and achieve TFD, we propose the Transparent Information Fusion Network (TIFN), a novel paradigm for transparent information fusion in fault diagnosis. Central to the TIFN design are self-organized Neural-Symbolic Nodes (NSNs), which integrate expert domain knowledge into the network architecture, replacing traditional neural nodes. These NSNs fuse and organize diverse expert knowledge after training, bridging the gap between the strong learning capabilities of deep learning and the explainability of symbolic systems. This fusion allows for the integration of deep learning representations with expert knowledge during both training and inference. Within the TIFN, two key types of NSNs are introduced: Signal Operator Nodes (SONs), which utilize existing signal processing expertise to establish the Signal Operator Layer (SOL), and Feature Operator Nodes (FONs), which employ statistical knowledge to create the Feature Operator Layer (FOL). Specifically, SONs employ the wavelet filter operator (WFO) [29], Hilbert transform (HT), and Identity transformation as the prior operators with learnable parameters and structural gates. For the FONs, we employ several statistical features to form FOL [41]. By stacking multiple SOLs and FOL, the TIFN facilitates transparent, information-fused decision-making processes. The self-organized NSNs not only enhance model explainability but also ensure that the entire decision-making process is transparent and explainable, thereby increasing trust in the outputs of systems. By embedding domain knowledge throughout

Table 1
The nomenclature of abbreviations.

Abbreviation	Explanation	Abbreviation	Explanation
BF	Ball fault	OF	Outer race fault
CE	Cross entropy	PHM	Prognostic and Health Management
FOL	Feature operator layer	RCC	Ribbon cage crack
FON	Feature operator node	SOL	Signal operator layer
HT	Hilbert transform	SON	Signal operator node
I	Identity transformation	TFD	Transparent fault diagnosis
IF	Inner race fault	TIFN	Transparent information fusion network
IFD	Intelligent fault diagnosis	WFN	Wavelet filter node
NOR	Normal condition	XAI	Explainable artificial intelligence
NSN	Neural-symbolic node	XPM	Explainable predictive maintenance

Table 2
The nomenclature of symbols.

Symbol	Explanation	Symbol	Explanation
$\mathcal{X}$	Data space	$\mathcal{Y}$	Label space
h_θ	Hypothesis with learnable parameters	$P(X,Y)$	Joint distribution
$(X,Y)=\{(x_i,y_i)\}_{i=1}^n$	Samples from P(X,Y)	$\mathcal{L}$	Loss function
$\mathcal{E}$	Expectation	R	Empirical risk
$\circ$	Composition	$\mathcal{H}$	Hypothesis space
$\rightarrow$	Mapping	$\mathcal{F}$	Feature space/Fourier transform
ϕ	Signal operator/SON	$\mathcal{F}^{-1}$	Inverse Fourier transform
τ	Feature operator/FON	Φ	The space of ϕ
σ	Decision operator	$\mathcal{T}$	The space of τ
θ	Learnable parameter	Σ	The space of σ
μ_c	Mean of channel signal	W	Channel-wise gates
μ_f	Mean of batch features	σ_c	Std of channel signal
B	Batch dimension	σ_f	Std of batch features
L	Signal dimension/Layer	C	Channel dimension
$\mathcal{K}$	Wavelet filter operator	SN	Signal normalization
f_b	Center frequency	R_θ	Wavelet in frequency domain
I	Identity transformation	f_c	Bandwidth
F	Extracted features	S	Enhanced signals

the learning process, TIFN aligns both symbolic and deep learning representations, establishing a solid foundation for explainability in IFD. This is particularly significant for Industry 5.0, where human-centric, transparent, and explainable AI systems are essential for trust and collaboration between humans and intelligent systems.

The contributions of this paper can be summarized as follows:

- To address the challenge of developing a fully and intrinsically explainable structure for TFD models, a TIFN is introduced using self-organized and knowledge-informed neural-symbolic nodes.
- Instead of the traditional neural nodes of deep neural networks, the NSNs incorporate prior knowledge of signal processing and statistical feature to enable explainable constraints.
- By connecting the multi-channel NSNs with learnable gates, intrinsically explainable SOLs and FOL are constructed to enhance the fault signature. Stacking multiple SOLs, a FOL and a softmax classifier ensures that all components of TIFN are comprehensible.
- Two case studies with different sensors are conducted to validate the information fusion ability of TIFN to achieve satisfactory diagnostic performance under a generalization task, shedding light on the potential of TIFN as a promising tool for making knowledge-informed industrial decisions.

The remainder of this paper is organized as follows: Section 2 presents the preliminary concepts and motivation behind TIFN. Section 3 details the proposed TIFN methodology. Section 4 provides the experimental results and analysis. Finally, Section 5 concludes the paper. The nomenclature of abbreviations and symbols is shown in Tables 1 and 2.

2. Preliminary

In this section, we present the issues with TFD and analyze the rationale behind the proposed TIFN method. To provide a clear explanation of TFD, we have included some essential notations below.

2.1. Intelligent fault diagnosis

In recent years, significant progress has been made in the development of IFD models. These models, based on statistical learning principles, can be represented as follows: $h_\theta : \mathcal{X} \rightarrow \mathcal{Y}$, where h_θ denotes the mapping function with learnable parameters, and $\mathcal{X}, \mathcal{Y}$ represent the spaces of collected signals and monitoring indices or fault types, respectively [42]. According to the Empirical Risk Minimization (ERM) principle [43], a training set is assumed to be sampled from a distribution $P(X,Y)$, represented as $\{(x_i,y_i)\}_{i=1}^n$, where x and y denote the input signals and labels respectively. To achieve optimal diagnostic results, ERM aims to minimize the expected risk by employing optimization algorithms targeted at a specific loss function $\mathcal{L} : Y \times Y \rightarrow [0,\infty)$ to construct well-fitted predictive models $h : X \rightarrow Y$ and their parameters θ. Given the challenge of accessing the complete signal distribution, the expected risk is approximated using empirical risk, estimated from the given data set. The optimal parameters θ^* and the corresponding model h^* are learned from the hypothesis space H by minimizing the empirical risk, as shown in Eqs. (1) and (2):

$$R(h) = \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}(y_i, h(x_i,\theta)), \tag{1}$$

$$h^* = \arg\min_{h\in H} R(h). \tag{2}$$

Once the model is trained, it can be employed to identify fault patterns in signals. The models have a hypothesis space, which includes all the decision functions as shown in Eq. (3):

$$\mathcal{H} = \{h | Y = h_\theta(X), \theta \in \mathcal{R}^n\}, \tag{3}$$

where the parameter θ can be learned from the dataset. Predictive models are often divided into two components, $h = \tau\circ\sigma$, where $\circ$ denotes the composition of models. Here, $\tau : \mathcal{X} \rightarrow \mathcal{F}$ represents the mapping from signal data to features, and $\sigma : \mathcal{F} \rightarrow \mathcal{Y}$ maps these features to fault patterns. Notably, neural networks, capable of learning

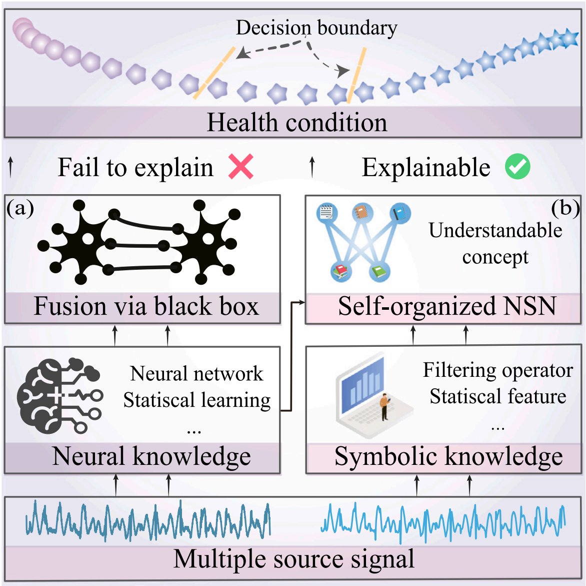

Fig. 2. Motivation for TIFN: (a) black box property of traditional IFD at the data-level and feature-level, (b) employing self-organized and understandable symbolic knowledge to substitute the black-box node.

features through nonlinear transformations, are particularly suited for constructing both τ and σ. This approach effectively incorporates empirical risk minimization into the process of IFD, leveraging both the theoretical framework and practical application of machine learning techniques to enhance the accuracy and reliability of fault detection systems.

2.2. Challenge: unexplainable learning in IFD

The unexplainable nature of IFD systems primarily stems from the assumption that the hypothesis space H, in which $h \in H$, is composed largely of neural network knowledge [44]. As illustrated in Fig. 2(a), these networks often function as "black boxes", making it challenging to interpret the mappings $\tau \in \mathcal{T}$ and $\sigma \in \Sigma$ that transform signal data into features and subsequently into fault patterns. Consequently, the decision-making processes of these models remain opaque, undermining the transparency and trustworthiness that are critical for deploying these systems in high-stakes industrial environments.

Moreover, modern industrial setups frequently utilize multiple types of sensors that generate diverse, multimodal data streams. The integration of this data through traditional fusion techniques often exacerbates the problem of explainability. Current fusion methods, which employ black-box approaches, obscure the operations performed during the fusion process, making it unclear how different data sources are combined to form the final diagnostic outcome.

2.3. Motivation: TFD with self-organized neural-symbolic nodes

Inspired by the integration of expert systems and neural networks, we aim to achieve a fully explainable diagnosis decision by training a neural network to obtain an explainable expert system. To this end and construct the TFD model, it is essential to establish a comprehensible learning space informed by expert knowledge. The traditional mapping $h_\theta : \mathcal{X} \to \mathcal{Y}$ can be decomposed into the following stages in this paper as shown in Eq. (4) to (6):

$$\phi : \mathcal{X} \to \mathcal{S}, \phi\left(x, \theta_\phi\right) = s, \tag{4}$$

$$\tau : \mathcal{S} \to \mathcal{F}, \tau\left(s, \theta_\tau\right) = f, \tag{5}$$

$$\sigma : \mathcal{F} \to \mathcal{Y}, \sigma\left(f, \theta_\sigma\right) = y, \tag{6}$$

where ϕ, τ, and σ represent the mappings from the input space to the enhanced signal space, feature space, and label space, respectively. The parameters θ_ϕ, θ_τ, and θ_σ are associated with these mappings. For instance, the WKN model [45] employs a learnable wavelet transform as the initial operator in the mapping ϕ in Eq. (4), followed by a CNN for subsequent layers. Similarly, the SincNet model [46] utilizes a learnable sinc function for the initial operator in the mapping ϕ in Eq. (4), with a CNN handling the later layers.

However, the black-box nature of neural networks, such as the mapping ϕ above, renders the diagnosis decision opaque and unexplainable. Therefore, we employ NSNs with explainable expert semantics instead of traditional neural nodes. These NSNs utilize understandable and learnable operators to represent expert symbolic knowledge, as illustrated in Fig. 2(b). Specifically, to achieve a self-organized TFD model, three modules are constructed using NSNs with gates w as described in Eqs. (7) to (9) [40]:

$$\phi : \mathcal{X} \to \mathcal{S}, \phi\left(x, w_\phi, \theta_\phi\right) = s, \tag{7}$$

$$\tau : \mathcal{S} \to \mathcal{F}, \tau\left(s, w_\tau, \theta_\tau\right) = f, \tag{8}$$

$$\sigma : \mathcal{F} \to \mathcal{Y}, \sigma\left(f, w_\sigma, \theta_\sigma\right) = y, \tag{9}$$

where ϕ, τ, σ are selected from the explainable operator spaces $\Phi, \mathcal{T}, \Sigma$, representing the sets of signal processing operators, statistical operators, and decision operators, respectively.

In each module, channel-wise gates w_ϕ, w_τ, w_σ are incorporated to determine the utilization of NSNs within the model. For gradient-based optimization, it is essential that both w and θ are differentiable. The optimization strategy not only aims to minimize empirical risk but also emphasizes reducing the L1-norm as an organizational penalty for the gates, thereby encouraging sparsity in model parameters.

The overall optimization objective is to find the model parameters that minimize both the empirical risk and the L1-norm of the gates as shown in Eq. (10):

$$\phi^*, \tau^*, \sigma^* = \min_{\phi\in\Phi,\tau\in\mathcal{T},\sigma\in\Sigma} \left(R(\phi\circ\tau\circ\sigma) + \lambda\|W\|_1\right), \tag{10}$$

where λ is a regularization parameter that balances the trade-off between empirical risk reduction and gate sparsity. Moreover, the operators are derived from an explainable learning space $\phi \in \Phi$, $\tau \in \mathcal{T}$, and $\sigma \in \Sigma$. Under this framework, achieving TFD becomes a promising endeavor.

3. Transparent information fusion network

To achieve TFD, as depicted in Fig. 3, the TIFN framework employs three types of modules after acquiring monitoring signals from multiple sensors. The first module consists of parameterized and self-organized SONs within the SOL, ensuring transparency in signal processing. Following the SOL, the physics-informed FOL extracts more effective health indices by incorporating prior statistical features. Finally, a classifier serves as a feature gate layer, mapping the extracted features to fault categories, thereby ensuring transparency in fault analysis for decision-makers. The following subsections provide a detailed explanation of these modules.

3.1. Signal operator layer

In our pursuit of the TFD model using multi-channel NSNs, we introduce the SOL as a fundamental layer, as illustrated in Fig. 3(b). Prior to being fed into the SOL, input signals $X \in \mathbb{R}^{1\times C_{in}\times L}$ must be standardized, which is commonly used in deep learning to normalize each channel independently as shown in Eq. (11):

$$\tilde{x} = SN(x_c) = \frac{x_c - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}}, \tag{11}$$

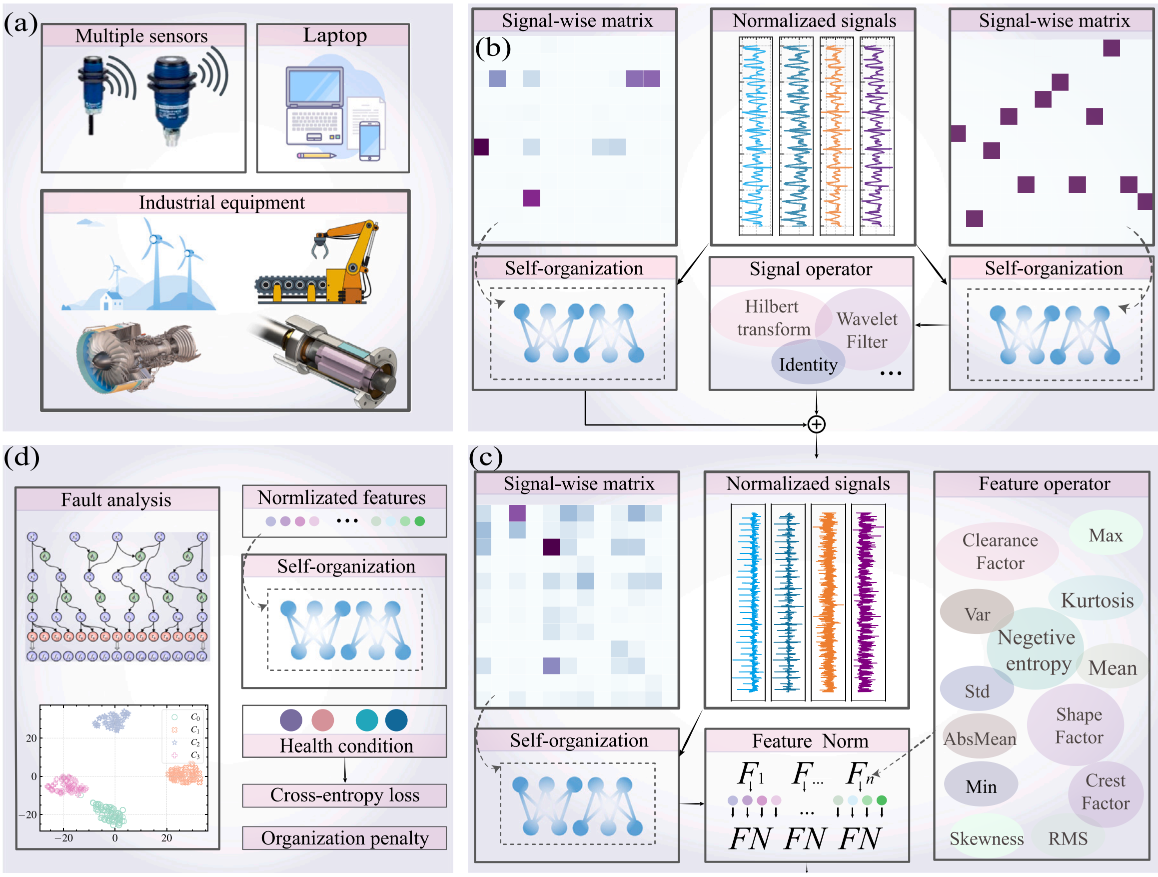

Fig. 3. Transparent information fusion network. (a) Signal acquisition, (b) signal operator layer, (c) feature operator layer, (d) linear classifier and fault analysis.

where $\mu_c = \frac{1}{L}\sum_{l=1}^{L} x_{c,l}$ represents the mean of X in the C channel, $\sigma_c^2 = \frac{1}{L}\sum_{l=1}^{L}(x_{c,l}-\mu_c)^2$ denotes the standard deviation of X in each channel, and L is the number of sample points of each channel. We then use a signal-wise matrix W_g to determine the interconnection between the operators in multiple channels as shown in Eq. (12):

$$W_g\tilde{X} = \begin{pmatrix} w_{11}\tilde{x}_1 & +\cdots & w_{1C_{in}}\tilde{x}_{C_{in}} \\ \vdots & \ddots & \vdots \\ w_{C_{out}1}\tilde{x}_1 & +\cdots & w_{C_{out}C_{in}}\tilde{x}_{C_{in}} \end{pmatrix}. \tag{12}$$

To achieve self-organizing functionality of explainable operators, the gate is scaled to address sparsity by applying a temperature-scaled column softmax function as shown in Eq. (13):

$$w'_{ij} = \frac{e^{w_{ij}/T}}{\sum_{i=1}^{C_{\text{out}}} e^{w_{ij}/T}}. \tag{13}$$

This approach enables the proposed method to autonomously extract expert systems from the learned explainable network, and adaptively fuse the enhanced signals and features within the expert system. Subsequently, the input to the explainable signal processing operator becomes the sparse signal $X_g = W'_g\tilde{X}$. The tensor is then divided into multiple parts along the channel dimension as a chunked tensor. Specifically, $X_g \in \mathbb{R}^{1\times C_{in}\times L}$ is split into $X_{g1}, X_{g\ldots}, X_{gN} \in \mathbb{R}^{1\times C_{in}//N\times L}$, with each split corresponding to a different signal processing operator as shown in Eq. (14):

$$X_i^{l+1} = \phi_i(X_{gi}^l), i = 1, 2, \ldots, N, X_i^{l+1} \in \mathbb{R}^{1\times C_{out}//N\times L}. \tag{14}$$

These signal processing operators ϕ_i are selected from the explainable operator space Φ, which is well-established and comprehensive, thereby eliminating the need for experts to manually construct explainable symbolic expert systems. Then, the X_i^{l+1} is concatenated into a new input of the next SOL. The signal-wise matrix W_s of residual connection, as given in Eq. (15), facilitates optimization. This is illustrated in Eq. (15) [29]:

$$X^{l+1} \leftarrow X^{l+1} + W_s X^l. \tag{15}$$

The representation enhances the health state-related information and can be used for subsequent monitoring purposes. The explainable operators ϕ employed in this study include multi-channel Hilbert Transform (HT), Wavelet Filter Operator (WFO) [29], and Identity Transformation. The function of the multi-channel setting is to facilitate the diverse fusion of enhanced signals and make the model scalable.

3.1.1. Hilbert transform

The Hilbert Transform (HT) is a crucial tool for extracting the envelope of a signal. For a given signal $x(t)$, the HT is defined as:

$$HT(t) = \frac{P}{\pi}\int_{-\infty}^{+\infty}\frac{x(\tau)}{t-\tau}\mathrm{d}\tau, \tag{16}$$

where P denotes the Cauchy principal value of the integral. Applying the Hilbert transform to the signal $x(t)$ results in its analytic signal $z(t)$, which consists of a real part $x(t)$ and an imaginary part $y(t)$ as shown in Eq. (17):

$$z(t) = x(t) + jy(t), \tag{17}$$

The magnitude of the analytic signal $|z(t)|$ is the envelope of the signal $x(t)$, which contains information about its amplitude modulation. The square of the envelope is then taken as the envelope spectrum.

Table 3
Features and their semantics.

No.	Feature name	Equations	No.	Feature name	Equations
0	Mean	$\frac{1}{N}\sum_{i=1}^{N} x_i$	7	Kurtosis	$\frac{\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^4}{\sigma^4}$
1	Std	$\sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^2}$	8	RMS	$\sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}$
2	Var	$\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^2$	9	CrestFactor	$\frac{\max(x)}{RMS(x)}$
3	Entropy	$-\sum_{i=1}^{N} p(x_i)\log p(x_i)$	10	ClearanceFactor	$\frac{\max(x)}{\frac{1}{N}\sum_{i=1}^{N}\lvert x_i\rvert}$
4	Max	$\max(x)$	11	Skewness	$\frac{\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^3}{\sigma^3}$
5	Min	$\min(x)$	12	ShapeFactor	$\frac{RMS(x)}{\frac{1}{N}\sum_{i=1}^{N}\lvert x_i\rvert}$
6	AbsMean	$\frac{1}{N}\sum_{i=1}^{N}\lvert x_i\rvert$	13	CrestFactorDelta	$\frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_{i+1}-x_i)^2}}{\frac{1}{N}\sum_{i=1}^{N}\lvert x_i\rvert}$

3.1.2. Wavelet filter operator

The WFO is a powerful tool for extracting frequency signatures, which can be formulated as shown in Eq. (18):

$$(\mathcal{K}(\theta)\mathbf{X}) = \mathcal{F}^{-1}\left(R_\theta(w)\cdot(\mathcal{F}\mathbf{X})\right), \tag{18}$$

where $\mathcal{F}$ denotes the Fourier transform, and $\mathcal{F}^{-1}$ denotes the inverse Fourier transform. The wavelet is choosen as Morlet, and its frequency domain representation as shown in Eq. (19):

$$R_\theta(w) = e^{-(\omega-2\pi f_c)^2/(4f_b^2)}, \tag{19}$$

where $\theta = \{f_b, f_c\}$ represents the parameterized center frequency and bandwidth, respectively.

3.1.3. Identity

The Identity operator simply outputs the same as the input, serving as a placeholder for convenient implementation and optimization as $I(x) = x$.

3.2. Feature operator layer

Following the signal processing layers, the signals from various channels are further weighted to assess the importance of different statistical operators. This process is managed by the FOL, which integrates multiple feature extraction modules to enhance the explainability and effectiveness of model. The FOL can be formulated as Eq. (20):

$$F = \mathcal{T}(W_\tau S). \tag{20}$$

Here, W_τ is the signal-wise matrix applied to the enhanced signal $S \in \mathbb{R}^{N\times C\times L}$ along each channel, transforming it into an intermediate representation that emphasizes the most relevant features. Consequently, the input signal $S \in \mathbb{R}^{N\times C\times L}$ is transformed into the output feature vector $F \in \mathbb{R}^{N\times(n\cdot C)\times 1}$ through the feature mapping function. The term $n\cdot C$ represents n statistical features across C channels, resulting in a total of $n\cdot C$ features. All of the $\tau \in \mathcal{T}$ are listed in Table 3.

To stabilize the model training, we propose the running feature normalization (RFN). During training, the mean μ and variance σ^2 of input features in the batch dimension are calculated to obtain statistical values for each channel as shown in Eq. (21):

$$\mu_f = \frac{1}{N}\sum_{i=1}^{N} f_i, \qquad \sigma_f^2 = \frac{1}{N}\sum_{i=1}^{N}(f_i-\mu_f)^2. \tag{21}$$

The running statistics are updated using an exponential moving average with a momentum parameter $\epsilon = 1e^{-6}$ according to Eq. (22):

$$\mu_{\text{running}} = (1-\epsilon)\mu_{\text{running}} + \epsilon\mu_f, \quad \sigma^2_{\text{running}} = (1-\epsilon)\sigma^2_{\text{running}} + \epsilon\sigma_f^2. \tag{22}$$

During the training phase, the input features are normalized using the batch statistics:

$$\hat{x} = \frac{x-\mu_f}{\sqrt{\sigma_f^2+\epsilon}}. \tag{23}$$

In the evaluation phase, the accumulated running statistics are employed for normalization:

$$\hat{x} = \frac{x-\mu_{\text{running}}}{\sqrt{\sigma^2_{\text{running}}+\epsilon}}. \tag{24}$$

This running feature normalization mechanism effectively reduces the scale variations across different features, thereby enhancing both the training stability and model performance.

3.3. Classifier and training strategy

For fault classification, a shallow softmax classifier processes the gated features to generate the final diagnostic decision:

$$P(Y|F) = \text{softmax}(W_\sigma F), \tag{25}$$

where W_σ denotes the learnable weights of the linear layer, and F represents the extracted feature representation. The model is trained using the standard cross-entropy (CE) loss function, formulated as:

$$\mathcal{L}(P(y), P_{W,\theta}(\tilde{y}|x)) = -\frac{1}{N}\sum_{i=1}^{N} P(y_i)\log P_{W,\theta}(\tilde{y}_i|x_i), \tag{26}$$

where W and θ represent the connection gates and WFO parameters, respectively. Following Occam's Razor principle [47], TIFN aims to learn the most parsimonious representation through sparse connections. This is achieved by adding an L1 regularization term:

$$\zeta(W) = \|W_{\phi,\tau,\sigma}\|_1 \tag{27}$$

The complete optimization objective, which is minimized using stochastic gradient descent [48], combines both the classification loss and sparsity constraint:

$$\left(W^*, \theta^*\right) = \underset{(W,\theta)}{\arg\min}(\mathcal{L}(W,\theta) + \beta\zeta(W)), \tag{28}$$

where β controls the trade-off between classification accuracy and model sparsity.

4. Cases study

4.1. Dataset setting

4.1.1. Dataset 1: vibration and piezoelectric signal

We used a self-powered condition monitoring dataset obtained using a piezoelectric energy harvester [49]. The dataset contains voltage signals of piezoelectric and vibration signals from four health conditions:

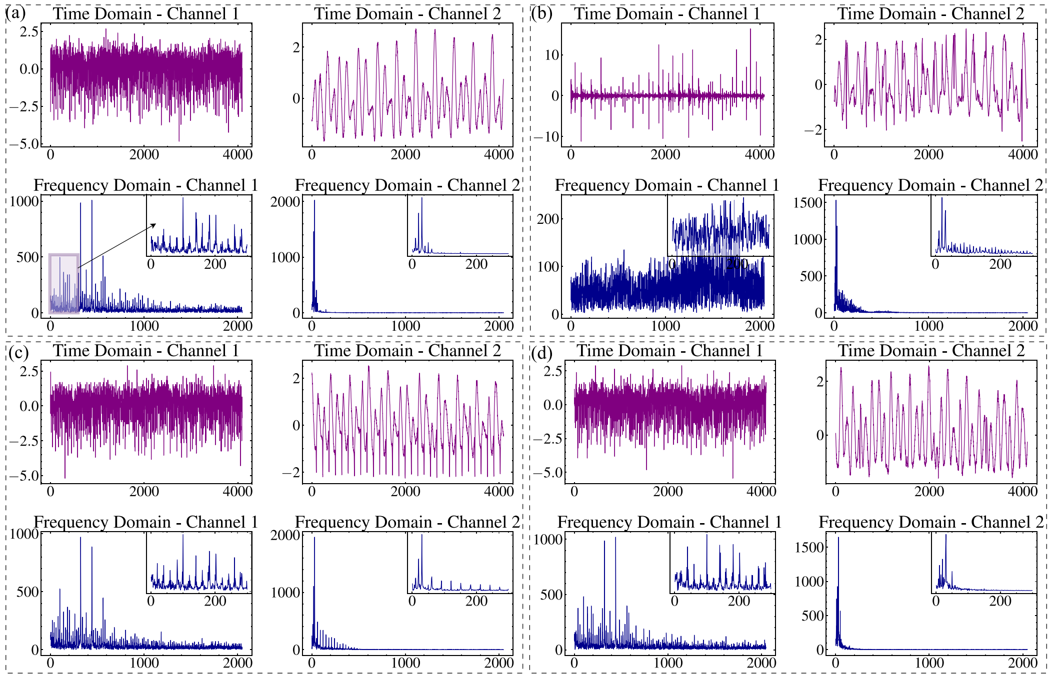

Fig. 4. Signals in Case 1. Channel 1 represents vibration signal, while Channel 2 represents voltage signal. (a) Nor, (b) IF, (c) BF, (d) OF.

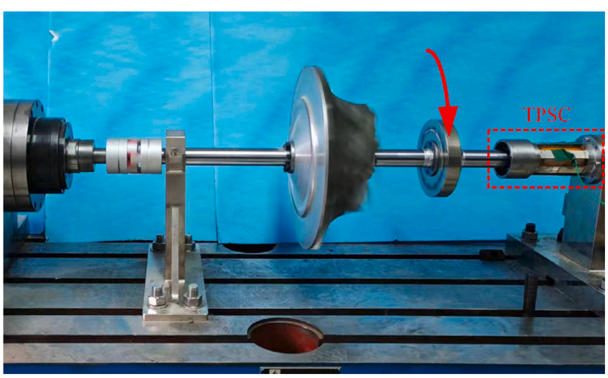

Fig. 5. Self-powered condition monitoring dataset obtained using a TSPC.

normal (Nor), inner race fault (IF), ball fault (BF), and outer race fault (OF) as shown in Fig. 4. The collected signals with 40960 Hz sampling frequency are downsampled to 4096 Hz and segmented at different time intervals to build 100 samples for each of the training set, validation set, and test set, respectively. Each sample consists of 4096 data points.

4.1.2. Dataset 2: Piezoelectric triboelectricity

We also employ a self-powered condition monitoring dataset obtained using a triboelectric-piezoelectric squirrel cage (TPSC) as illustrated in Fig. 5 of the study by Ref. [50]. The dataset encompasses voltage signals from piezoelectric and triboelectricity corresponding to five distinct health conditions of the aero-engine rotor system: Nor, OF, IF, BF, and ribbon cage crack (RCC). The signals, initially collected with a high sampling frequency of 40960 Hz, have been downsampled to 4096 Hz to enhance computational efficiency as shown in Fig. 6. These signals are then segmented at various time intervals, culminating in the creation of 100 samples for each of the training, validation, and test sets. Each sample consists of 4096 data points, providing a detailed representation of the status across a spectrum of operational states.

To evaluate the performance of the TIFN model, four distinct tasks were designed. Task **T1** utilized Dataset 1, while Task **T2** employed Dataset 2 to assess the performance of model. Additionally, Task **T1G** evaluated the generalization capability of the model using Dataset 1, which was trained at speeds of 1 Hz and 15 Hz and tested at a speed of 10 Hz. Task **T2S** examined the performance of model with a limited sample size using Dataset 2. All diagnostic models were implemented in Python 3.8 and PyTorch 1.12, with five different random seeds to mitigate the effects of randomness. The hyperparameters used for training the proposed TIFN model are as follows: the learning rate is set to 0.001, the batch size is 64, the number of epochs is 300, model scale is set by 4, and the weight decay is 0.0001. The model training was conducted on a GeForce RTX 4090 GPU.

4.2. Compared methods

To evaluate the effectiveness, model explainability, and diagnostic performance of our method, we compared it to several IFD methods with different parameters as shown in Table 4. One of these methods is ResNet fashion method [51], a well-known convolutional neural network in multiple domains including IFD [52]. As a neural fashion backbone, we chose the ResNet-18 version as M1 for comparison. Additionally, we also evaluated the Sincnet [46] as M2 and the WKN model [45] as M3, which use sinc filters and continue Morlet wavelet convolution as the expert knowledge in the ResNet. Furthermore, the MWA-CNN [53] utilized the Discrete Wavelet Attention Layer was also chosen as M4.

4.3. Diagnosis results

Table 4 summarizes the diagnosis accuracy of different IFD methods, comparing their model parameters, explainability, and performance across various tasks. The following analysis examines the results from different perspectives.

4.3.1. Parameters and explainability

TIFN has the fewest parameters, indicating a lighter model. Its full explainability means every decision made by the model can be understood and traced, which is crucial for applications requiring transparency and reliability. ResNet has a significantly higher number of parameters compared to TIFN. The lack of explainability makes it

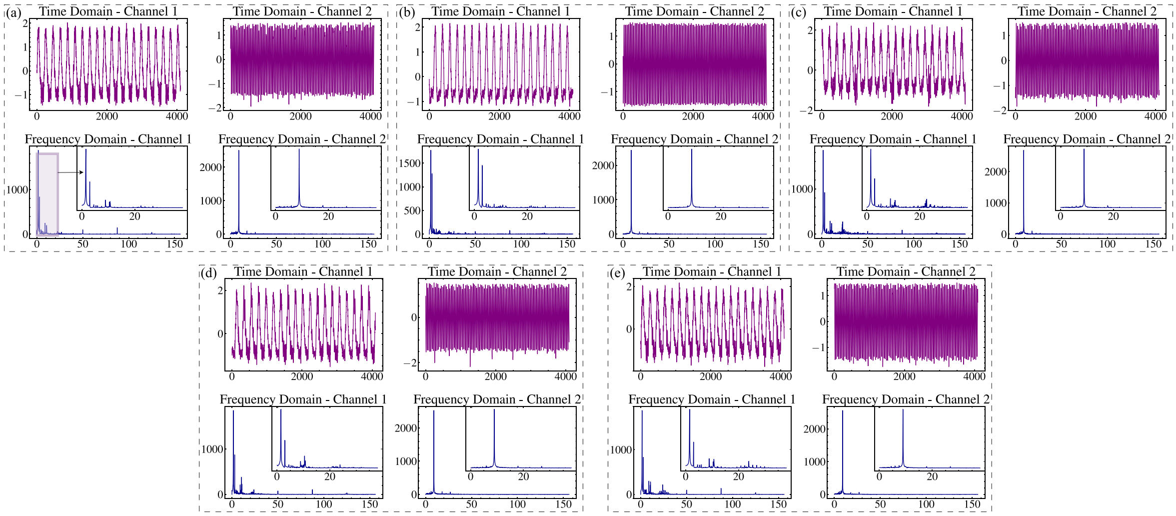

Fig. 6. Signals in Case 1. Channel 1 represents triboelectric signal, while Channel 2 represents piezoelectric signal. (a) Nor, (b) OF, (c) IF, (d) BF, (e) RCC.

Table 4
Diagnosis accuracy of T1, T2 and T1G.

Methods	Parameters	explainability	T1 (%)	T2(%)	T1G (%)
TIFN	**21.94k**	Fully	100 ± 0	**95.40** ± **1.50**	**99.04** ± **0.46**
M1	384.64k	/	100 ± 0	93.56 ± 4.78	52.38 ± 4.17
M2	384.55k	Partial	100 ± 0	87.58 ± 7.11	50.34 ± 23.34
M3	384.55k	Partial	100 ± 0	85.86 ± 11.86	61.86 ± 19.38
M4	87.00k	Partial	97.61 ± 4.21	35.17 ± 4.56	96.05 ± 2.77

a black-box model, which can be a disadvantage in scenarios where understanding the decision-making process is critical. SincNet also has a high number of parameters similar to ResNet. The use of parameterized Sinc functions in the first layer offers some explainability, but the overall model remains largely a black-box. WKN integrates wavelet transformations for enhanced feature extraction, providing partial explainability. Despite the similar parameter count to ResNet and SincNet, its use of domain-specific knowledge can offer insights into certain decisions made by the model. MWA-CNN has a moderate number of parameters and utilizes Discrete Wavelet Attention Layer, providing some explainability. It balances complexity and explainability better than ResNet and SincNet.

4.3.2. GPU acceleration

In contrast to traditional symbolic diagnosis systems with expert knowledge, TIFN leverages GPU to expedite self-organization processes. The comparison in Table 5 illustrates the computational times and efficiency of various operations with and without GPU acceleration. With GPU active, the role of CPU primarily involves management and coordination, leading to a significant decrease in its workload. Consequently, CPU processing times are notably reduced as the GPU handles most of the self-organization of operators. The final column demonstrates the percentage decrease in processing time when utilizing the GPU as opposed to CPU-only execution. For instance, enabling GPU for the signal operator layer can result in approximately a tenfold time saving, significantly enhancing the efficiency of information fusion and self-organization of knowledge.

4.3.3. TIFN achieves satisfactory performance in fundamental tasks T1 and T2

All methods achieve near-perfect performance on T1, indicating their suitability for basic fault diagnosis tasks. However, the performance varies significantly in T2. TIFN outperforms all other models, demonstrating its robustness and adaptability to more complex scenarios. While ResNet and SincNet also perform well, they exhibit less consistency compared to TIFN. WKN and MWA-CNN show significantly unsatisfactory performance, indicating potential weaknesses in handling this particular task.

4.3.4. TIFN can generalize well in the T1G task

T1G task evaluates the generalization capability of the models. TIFN demonstrates superior performance, indicating its ability to generalize well across different conditions. MWA-CNN also shows strong generalization, though it struggles with T2. ResNet and SincNet perform unsatisfactory in terms of generalization, highlighting a potential overfitting issue. WKN shows moderate generalization but is still outperformed by TIFN and MWA-CNN.

4.3.5. TIFN performs well in limited sample scenarios of the T2S task due to the knowledge-informed NSNs

The experimental results in Table 6 demonstrate the robust performance of TIFN across varying sample sizes. Starting with a competitive accuracy of 64.22% at N=5 samples, TIFN exhibits consistent improvement as the sample size increases, ultimately achieving a remarkable 96.10% accuracy at N=30. In comparison, the baseline methods show varying degrees of effectiveness: ResNet demonstrates gradual improvement from 57.47% to 86.21%, while SincNet maintains moderate performance ranging from 67.38% to 87.59%. WKN initially excels with 70.83% accuracy at N=5 but plateaus at 89.66% with N=30. Notably, MWA-CNN displays inconsistent performance across sample sizes, which we attribute to its explainable optimization for vibration signals potentially interfering with the information fusion process when processing speed signals. These comprehensive results substantiate superior performance of TIFN, particularly in scenarios with larger sample sizes, while maintaining competitive accuracy even in few-shot learning conditions. The consistent improvement in accuracy with increasing sample size demonstrates the robust learning capabilities and effective information fusion mechanism.

Table 5
A comparison of computational times and efficiency for different operations with and without GPU acceleration.

Operation description	CPU time with GPU (s)	CUDA time with GPU (s)	CPU time Only (s)	Time saved (%)
Training Epoch	20.38200	0.02831	42.06000	51.47
Training Batch	0.05199	0.00181	0.18930	71.58
Optimization Step	0.05176	0.00181	0.18906	71.66
Backward Pass	0.02359	0.00000	0.09532	75.25
Training Process	0.01892	0.00181	0.08911	76.74
Signal operator Layers	0.01363	0.00163	0.10335	85.24
Feature operator Layer	0.00307	0.00055	0.01958	81.48
Signal operator Layer 1	0.00298	0.00021	0.03009	89.41
Signal operator Layer 2	0.00239	0.00028	0.01958	86.36
Signal operator Layer 3	0.00231	0.00028	0.01898	86.32
Signal operator Layer 4	0.00225	0.00028	0.01737	85.45

Table 6
Diagnosis accuracy of limited sample task from N=5 to 30 (%)

Methods	N=5	N=10	N=15	N=20	N=25	N=30
TIFN	64.22	**82.32**	82.40	**89.66**	**95.98**	**96.10**
M1	57.47	60.06	66.24	71.26	71.98	86.21
M2	67.38	73.99	83.33	88.36	88.51	87.59
M3	**70.83**	77.87	**86.06**	87.21	87.21	89.66
M4	57.33	20.69	64.66	40.52	20.69	54.74

4.4. Ablation study

To further understand the contributions of different components in the TIFN model, we conducted an ablation study. The settings can be seen in the provided Fig. 7, which illustrates the results of the ablation study conducted to evaluate the effects of different components in the model. Each sub-figure presents the performance under varying configurations of learning rates for gated weights (horizontal axis, λ_g) and filter parameters (vertical axis, λ_p). The color scale from blue to magenta represents validation accuracy, where blue indicates lower performance and magenta indicates higher performance.

Panel (a) represents the performance of the model when all components are included, serving as the baseline for comparison. The model shows a relatively balanced performance across different learning rates, with optimal performance in the mid-range of both λ_g and λ_p. This suggests that the full model is robust to variations in these parameters. For the configuration using only the Identity component in Fig. 7(b), omitting other SONs, performance significantly drops compared to the full model, especially at higher values of λ_g and λ_p. This indicates that the Identity component alone is insufficient for effective learning. When the feature set is limited to the Mean, as shown in Fig. 7(c), the model achieves lower performance in certain regions, suggesting that only the Mean component contributes less. Fig. 7(d) excludes the residual connections from the model. The absence of residual connections leads to significant performance degradation across all learning rates. This highlights the importance of residual connections in stabilizing the learning process and enhancing performance. Additionally, the absence of the HT or WFO leads to decreased performance, especially at higher learning rates in Fig. 7(e) and (f).

Moreover, we also employed the Ricker wavelet filter (RWF) [54], Laplace wavelet filter (LWF) [45], and Chirplet wavelet filter (CWF) [55] as replacements for the original Morlet wavelet, with results shown in Fig. 7(g), (h), and (i). It can be observed that while the Ricker and Chirplet wavelets achieve promising training results under certain λ_g and λ_p settings, they lack the stability of the Morlet wavelet in this context. Additionally, the Laplace wavelet fails to capture the features effectively.

4.5. Feature visualization

Figs. 8 and 9 display the T-SNE (t-distributed Stochastic Neighbor Embedding) feature visualization for two cases. T-SNE is a powerful tool for visualizing high-dimensional data by projecting it into a lower-dimensional space while preserving the structure and relationships between data points. Here, the features extracted by different models are visualized to assess their effectiveness in clustering and distinguishing between different fault conditions.

The TIFN model shows distinct and well-separated clusters for different fault conditions in T1. This indicates that TIFN can effectively extract and distinguish relevant features, making it easier to diagnose faults. ResNet also shows distinct clusters, but there is more overlap between some fault conditions compared to TIFN. SincNet displays fairly well-separated clusters, though some overlap is noticeable. WKN shows clear clusters, but there are significant overlaps, particularly in some fault conditions. MWA-CNN shows distinct clusters with some overlap, similar to SincNet. However, in T2, TIFN continues to show very distinct and well-separated clusters, reaffirming its robustness in feature extraction and fault distinction, whereas M4 seems to struggle to handle the multi-source signal fusion task.

4.6. Transparent analysis

To elaborate on the explainability of the model, node visualization from the TIFN of Dataset 1 is presented with multiple interconnected layers in Fig. 10. In TIFN, the principle of Occam's Razor is adhered to, which demands the extraction of the most compact representation. Therefore, an iterative pruning strategy is employed to achieve a simplified model. This is done by removing the lowest connected weight and its associated symbolic operator. For more details, the reader can refer to [40]. The transparency is determined by gates controlled by the temperature-scaled softmax function in Eq. (13), illustrating the gated connections within the SOL and FOL frameworks. Through the processes of learning and sparsification, the TIFN refines its focus on key operators. For example, s_6 is generated from the vibration signal after applying the Hilbert transform and wavelet filter. It is subsequently utilized for feature extraction at the feature level, focusing on mean, entropy, and skewness. Additionally, x_7^1 integrates signals of two different modalities at the data level, employs two distinct wavelet filters ϕ, and ultimately captures the standard deviation for feature representation. Finally, the representative feature can be obtained by the self-organized decision route, which is the key feature for fault diagnosis. As shown in Fig. 11, the representative features f_1 (variance), f_2 (RMS), f_7 (minimum), and f_{12} (kurtosis) are reordered and highlighted, demonstrating the significant discriminative capability for distinguishing between different health conditions. As a result, engineers can use these visualizations to understand which parts and features of the signal are most affected by faults, while maintenance teams can focus on the key NSNs with specific self-organized decision routes identified by the sparse networks, ensuring efficient and targeted fault diagnosis.

5. Conclusion and future work

Our study presented a novel approach to achieving Transparent Fault Diagnosis (TFD). We proposed the Transparent Information Fusion Network (TIFN), which utilized self-organized and knowledge-informed Neuro-Symbolic Nodes (NSNs) instead of traditional neural

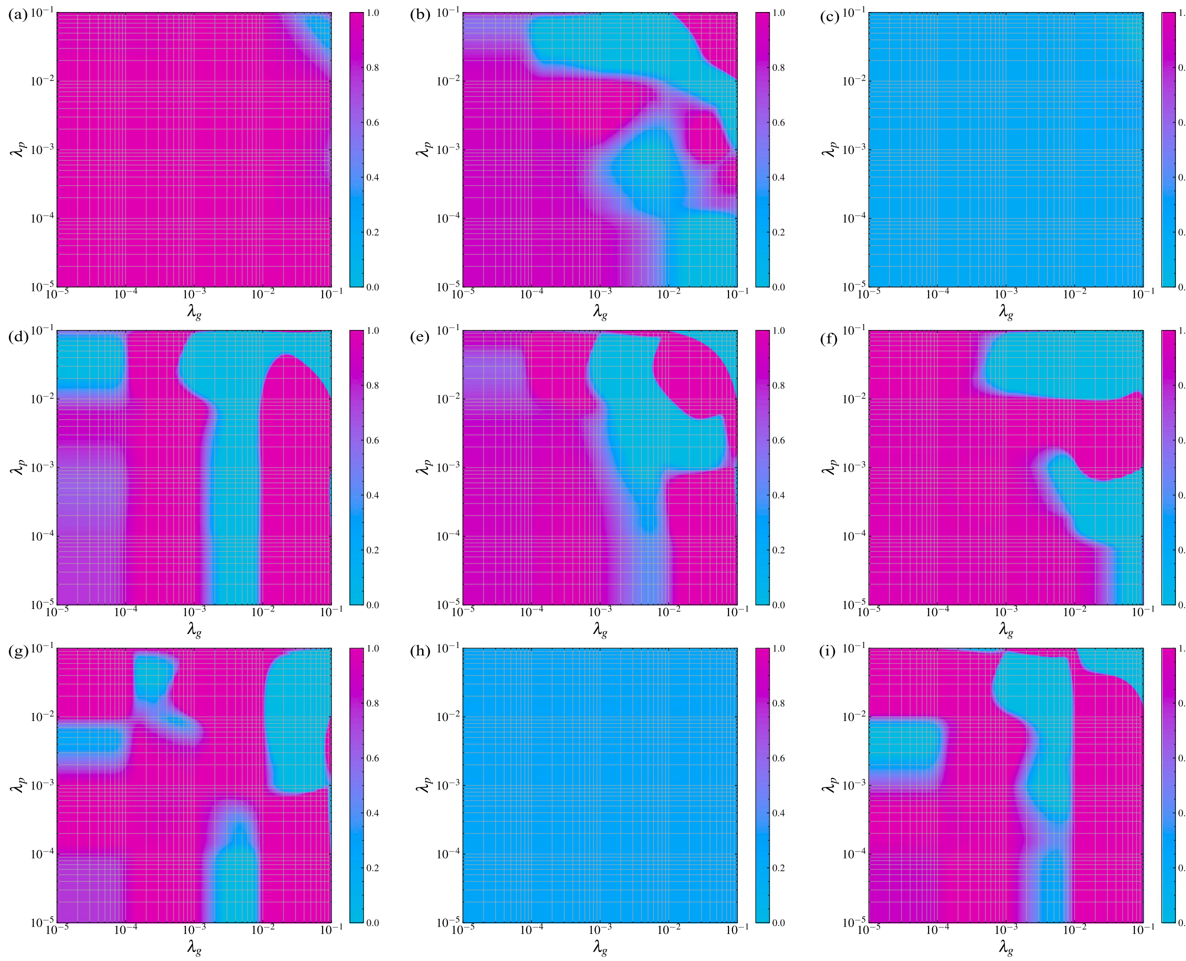

Fig. 7. Ablation study results. (a) All components included, (b) only Identity component, (c) only Mean feature, (d) without residual connection, (e) without HT component, (f) without WFO component, (g) using CWF instead of WFO, (h) using LWF instead of WFO, (i) using RWF instead of WFO.

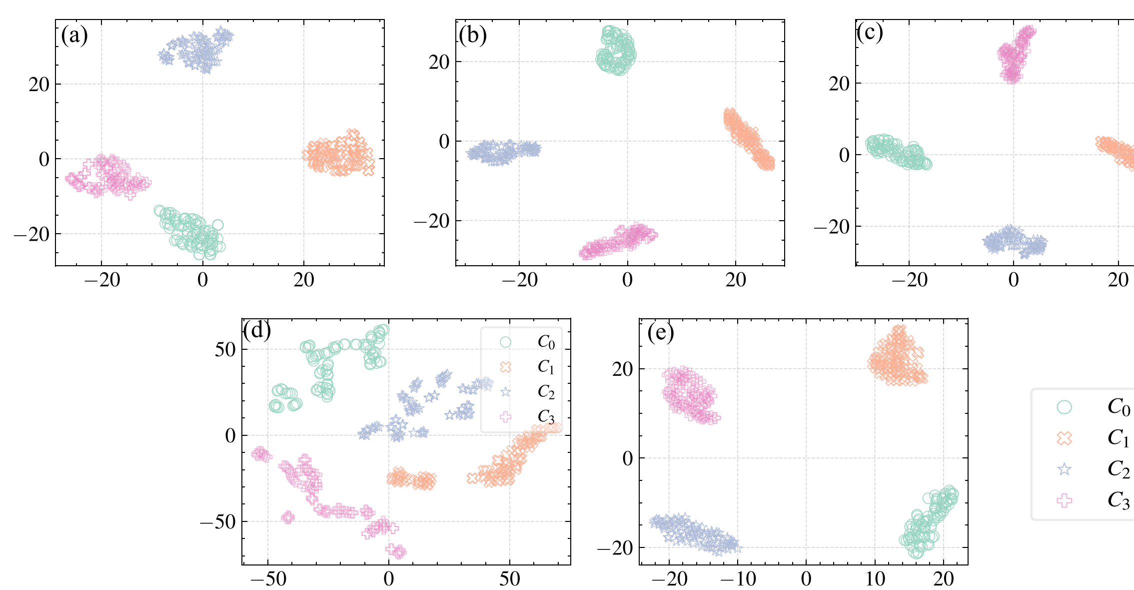

Fig. 8. Feature Visualization for Case 1: (a) TIFN, (b) M1, (c) M2, (d) M3, (e) M4.

nodes. The NSN integrated prior knowledge of signal processing and statistical features, applying explainable constraints. By connecting multiple NSNs with gates, we constructed an intrinsically explainable Signal Operator Layer (SOL) and Feature Operator Layer (FOL), which together enhanced fault signatures. Stacking multiple SOLs along with the FOL and a softmax classifier resulted in the comprehensible TIFN. Two case studies validated the ability to achieve satisfactory diagnostic performance under generalization tasks and limited sample conditions,

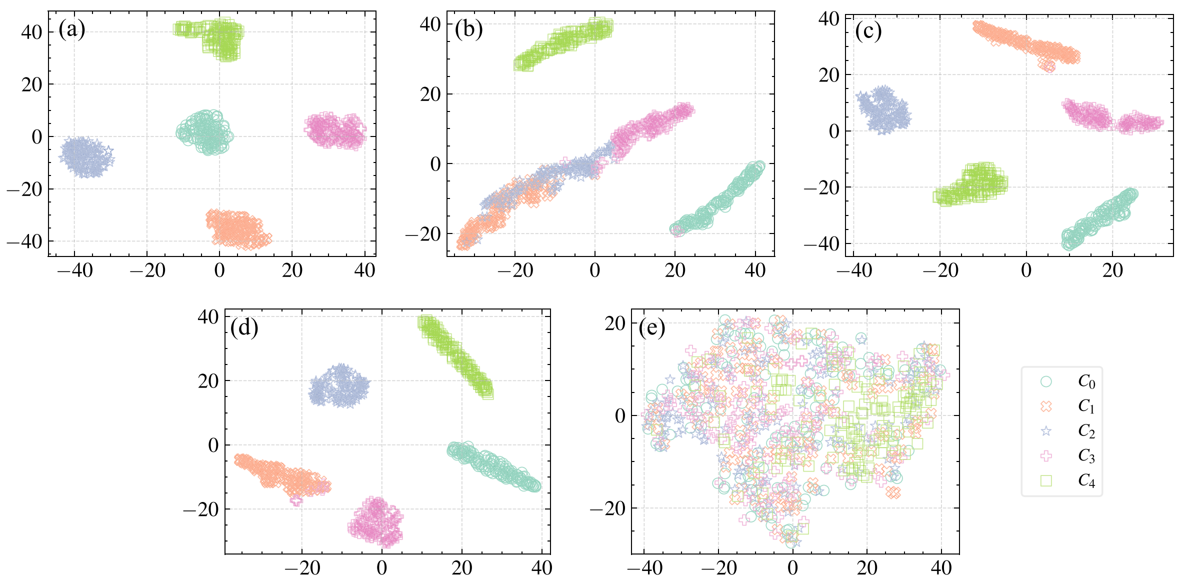

Fig. 9. Feature Visualization for Case 2: (a) TIFN, (b) M1, (c) M2, (d) M3, (e) M4.

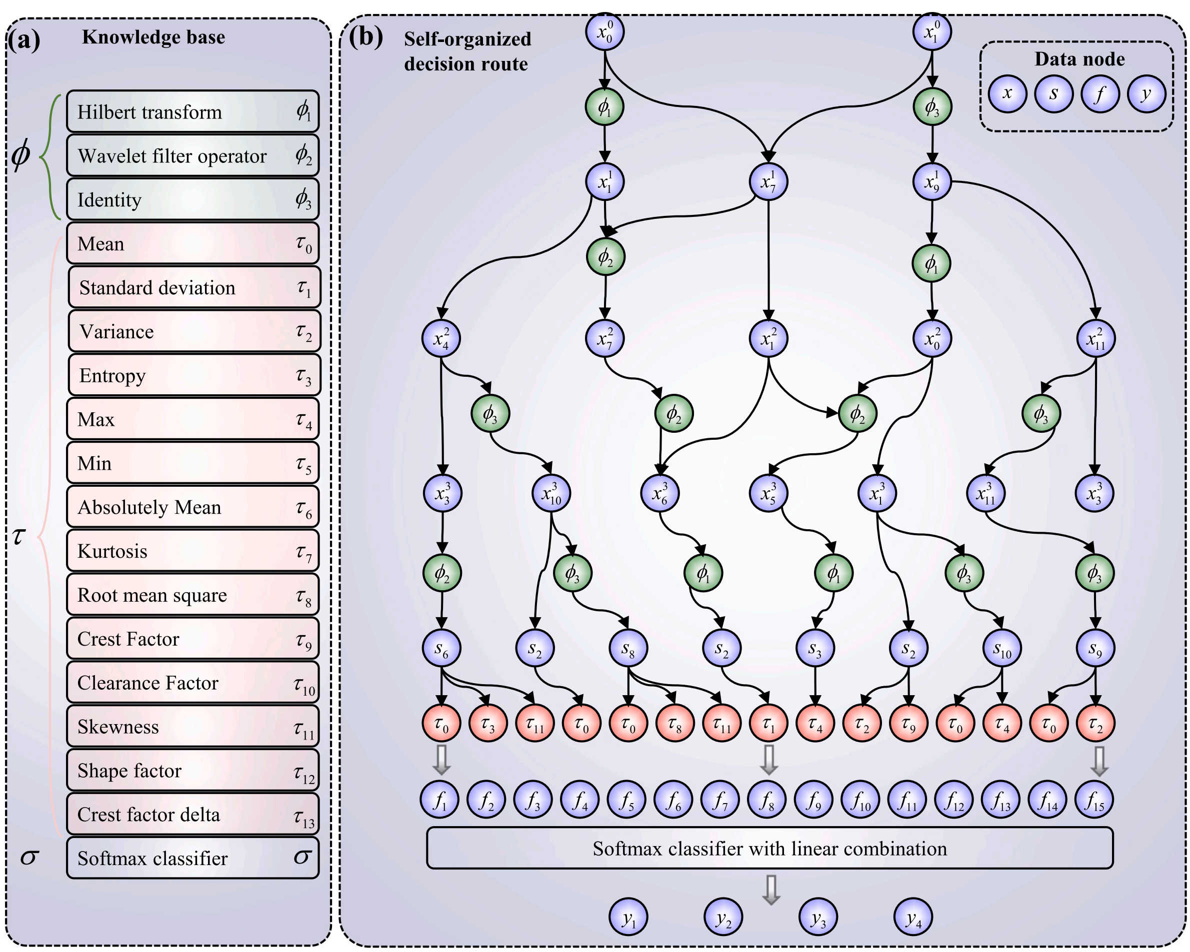

Fig. 10. Self-organization of the TIFN. (a) knowledge base, (b) self-organized decision route.

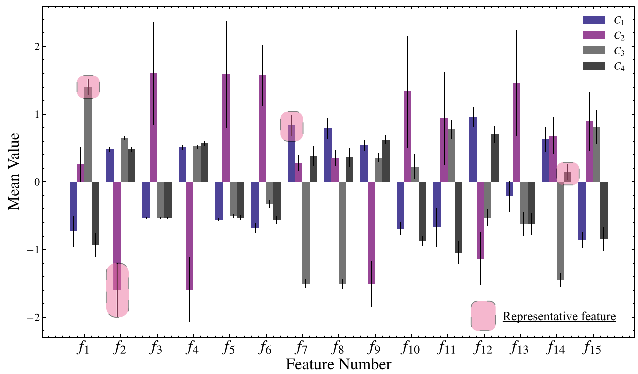

Fig. 11. Representative features, with each feature bar displaying the mean and standard deviation for four different health conditions $C_{1,2,3,4}$. The representative features f_1 (variance), f_2 (RMS), f_7 (minimum), and f_{12} (kurtosis) are highlighted.

highlighting its potential as a valuable tool for knowledge-informed industrial decision-making with information fusion.

Our future work will focus on enhancing the explainability of neuro-symbolic nodes and integrating additional symbols to improve transparent fault diagnosis in industrial applications. It is also noteworthy that no activation functions are currently employed in the SOL and FOL. Future research could explore the design of explainable activation functions. Additionally, developing a versatile wavelet self-organization method holds promise for broader applications. The training strategy should also address parameter initialization and stabilization, particularly when scaling up the model.

CRediT authorship contribution statement

Qi Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lichang Qin:** Writing – original draft, Resources, Project administration, Data curation. **Haifeng Xu:** Writing – original draft, Formal analysis, Data curation. **Qijian Lin:** Writing – original draft, Data curation. **Zhaoye Qin:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization, Funding acquisition. **Fulei Chu:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GPT-4o in order to polish the language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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